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RESEARCH ARTICLE

Node-Based Graph Convolutional Network With SLIC Method for Breast Cancer Ultrasound Images Classification

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ABSTRACT This research presents a novel node-based Graph Convolutional Network (GCN) approach for the classification of breast cancer from ultrasound images. Utilizing the Simple Linear Iterative Clustering (SLIC) algorithm, superpixels are segmented and treated as nodes and connected based on color similarity and spatial proximity. This enables the graph data capturing both long-range and short-range dependencies among image features. The graph-structured data is processed through a multi-layer GCN, which effectively aggregates and learns from complex node interrelations to classify the nodes as benign, malignant, or normal. In the final stage, a voting rule is applied to the node classification in order to perform the final image-level classification. Overall, the proposed framework yielded impressive findings on two distinct datasets, Breast Ultrasound Images Dataset (BUSI) and Breast Ultrasound Image dataset from Brazil (BUSBRA). Different numbers of connections between nodes are evaluated to indicate the importance of hidden insight interconnections in the medical imaging domain. The results give an accuracy of 98.73% for the BUSI dataset and 98.40% in the case of the BUSBRA dataset, indicating superior performance compared to conventional and other advanced methodologies. This highlights the potential of employing node-based GCNs for breast cancer detection using ultrasound imagery in clinical applications.

INDEX TERMS Breast cancer ultrasound, deep learning, image classification, simple linear iterative clustering, graph convolutional network.

I. INTRODUCTION

Breast cancer is still considered as a worldwide health concern, which continues to be one of the most frequent cancers in females. The report [1] shows approximately 250,000 new cases being diagnosed in the United States during the year 2017 alone, and the incidence of suffering patients from this cancer is on the rise, as the evidence provided in [2]. Furthermore, there will be an expected increase in this type of cancer by the year 2040, with over 3 million new cases and about 1 million fatalities annually [3]. Therefore, early diagnosis of breast cancer plays an important role in not only enhancing the effectiveness of treatment but also improving the survival rate of patients [4]. Additionally, early identification may pave the way for a more extensive variety of treatments, some of which may be less invasive and more targeted. This may significantly reduce the physical and emotional impact of breast cancer treatment on patients. The process of screening for cancers requires the use of diagnostic techniques in order to identify most breast cancers in the beginning stages. By investigating various indicators, radiologists are able to identify these cancers by assessing the mass and calcification regions [5], [6]. The reduction by up to 63% in mortality rates has been proven in [7] by the utilization of screening mammography. Nevertheless, identifying these indicators is a highly skilled task that frequently carries a 20% margin of error [8].

Despite the fact that mammography has historically been the gold standard, particularly in high-income nations, ultrasound screening has become a powerful supplemental

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and alternative screening method for breast cancer in some cases. Unlike mammography, ultrasound is considered a safer option without ionizing radiation for those requiring more frequent screening because of the high-risk or dense breast [9]. This allows applying in the case of pregnant women [10]. Besides, this screening method has a high sensitivity in some dense breast tissue patients, where mammography might miss cancers due to the dense tissue obscuring the tumors [11]. Thus, ultrasonography has emerged as a widely employed technique in the fields of clinical breast cancer monitoring and identification [12]. Nevertheless, due to elevated sensitivity, the ultrasound machine is vulnerable to the influences of different tissues and the surrounding environment. This leads to a substantial quantity of inherent noise, low contrast, and uncertain contours [13], while the research [14] reported that the ability of conventional Computer-Aided Design (CAD) systems in correctly identifying cases only reaches 30%.

Deep Learning (DL) has demonstrated remarkable efficacy in analyzing complicated medical imaging. Convolutional Neural Network (CNN) is known as a powerful type of deep learning architecture, which is able to analyze visual imagery by employing layers of processing to identify patterns and features effectively. Several pre-trained models, including ResNet [15], VGG16 [16], MobileNet [17], EfficientNet [18], etc., have been introduced and developed to address the complicated task of image analysis. There is a growing trend of developing custom advanced models to work effectively with different medical imaging techniques. For instance, the specialized lightweight network has been designed to handle Computed Tomography (CT) scans successfully [19]. A notable example proposed an enhanced architecture from the baseline to deal with ultrasound images [20], while the study [21] summarized the custom models for ultrasound comprehensively with their advantages and challenges, giving rationale ideas for the further development trend. Advancements in MRI analysis are exemplified by the improved version of Multimodal Lightweight XGBoost framework [22], and the modified YOLO architectures are leveraged to increase the ability of classification and segmentation [23]. Additionally, the fusion of different imaging techniques can also be considered, which received better performance in assessment [24]. Specifically, various advanced models have been developed to work effectively with different medical imaging techniques. These models have been successfully utilized in the field of breast cancer diagnosis [25], [26], [27], [28], [29], by demonstrating notable performances.

Nevertheless, CNNs that utilize spatial hierarchies are particularly effective at processing grid-like data, such as images. Irregular data structures and complex relationships frequently present obstacles for these networks [30]. This can be explained by the fact that CNN-based models do not analyze correlations between visual data at a finer level and are only strong when working with short-range pixels, while capturing the global and local context may play an essential role in working with medical images. Graph Convolutional Networks (GCNs) operate with convolutional processes on data represented as graphs, where nodes correspond to elements [31]. Their ability is able to model complex dependencies and interactions, which are difficult to capture with CNNs. Especially, this problem may worsen in the case that the spatial relationships are irregular or when the global context and relational information among elements become crucial [32]. Recently, GCN became an option to tackle these limitations by conceptualizing the image data as graphs. Another important point is that GCNs can leverage much more additional information and prior knowledge through the use of graph edges to provide superior feature extraction and representation learning targeted toward domain-specific applications [33], [34]. Thus, this offers a novel approach to medical image analysis, which considers nodes as medical findings or pixel data [35]. Besides, the relationships between these entities are treated as edges. This structure allows GCNs to gain a deeper insight into complex medical datasets.

Therefore, this study presents an advanced framework for breast cancer classification via ultrasound images using a Graph Convolutional Network (GCN) based on node classification of superpixels segmented using the Simple Linear Iterative Clustering (SLIC) algorithm. Expanding upon prior research [36] and [37], our approach considers not only the connections between adjacent superpixels, but also the similarities in characteristics. Thus, each superpixel is treated as a node and connected based on color similarity and spatial proximity. Thanks to this configuration, the framework is able to work with long-range dependencies and acquire incredibly potential feature representations. This may bring advantages in analyzing medical imaging to obtain the hidden correlations of pixels, resulting in boosting performance for identifying tumors. Node labels are determined by the overlap of superpixel centroids with annotated regions in mask images. The graph comprising nodes and edges is processed through a GCN. Based on the classified nodes, the final image classification derives from a voting rule applied to the node classifications, thus integrating local node predictions into a comprehensive image-level diagnosis. The effectiveness of this approach is assessed quantitatively through various performance metrics, which demonstrate its superiority over conventional and other advanced methodologies. This demonstrates the practicality of node-based GCN in improving classification procedures.

Overall, the primary contributions of this study can be summarized as follows:

- Introducing a node-based GCN approach to enhance the classification of breast cancer from breast ultrasound imaging.
- Proposing advanced implementation of node-based transformation from image to graph data, which effectively captures the global and local context of the image by balancing the features in terms of spatial and property context.

- Incorporating the node classification to provide the entire image classification, which is able to highlight the abnormal region. This enhances the interpretability of the framework being offered.
- Finally, validating our proposed work on two distinct datasets, namely BUSI and BUSBRA, which have noteworthy enhancements in performance.

The remainder of this manuscript is structured as follows. Section II offers an exhaustive review of the previous research endeavors concerning breast cancer classification. Section III provides a detailed description of the proposed node-based GCN methodology. The experimental outcomes and subsequent discussions are detailed in Section IV of the paper, while Section V of the paper contains the conclusion and potential directions for future research.

II. RELATED WORK

A large amount of research has been carried out to implement automated breast cancer classification to support the diagnosis process thanks to the advancement in computer vision. Initially, deep CNN is applied to perform end-to-end classification, which does not require manually extracting the image data and can utilize the advantages of a large amount of data. Miron et al. [38] focus on the preprocessing and optimal feature extraction of regions of interest (ROIs) from ultrasound images, while a simple network is integrated to work as a classifier - reaching 86% validation accuracy. By extracting the deep features and classification, Yu et al. [39] employ the pre-trained model Inception-V3 to improve diagnostic accuracy up to 85%, while the study [40] uses the ResNet101model as a primary backbone for classification, giving 92.3% in terms of testing accuracy.

To improve the performance of the identification process, the joint model is deemed to integrate the formidable characteristics of each individual model. Indeed, Ayana et al. [41] address the critical challenge of limited training datasets in medical imaging by utilizing a multistage transfer learning (MSTL) approach. The experiments yielded noteworthy enhancements in the accuracy of the test set, attaining 99% on the Mendeley dataset and 98.7% on the MT-Small-Dataset. The ensemble architecture proposed by Zhuang et al. [42] integrates three pre-trained models, fuzzy enhancement and bilateral filtering, which help to clarify and enhance the image quality of ultrasound scans for classification. Thus, the accuracy percentage of 95.48% is proven by the given approach. Besides, the ResNet50 pre-trained model is applied for feature extraction in conjunction with Binary Gray Wolf Optimization (BGWO), forming a hybrid approach as Khanna et al. [43] proposed. This optimization algorithm helps to select the essential features before feeding into the Support Vector Machine (SVM) classifier, which indicates a classification accuracy of 84.9% and AUC of 0.97. The optimization algorithm for feature selection is also utilized by Ahila et al. [44] in the case of breast cancer classification using ultrasound. Instead of using CNN, the authors suggest employing Gray Wolf Optimization (GWO)-tuned Wavelet Neural Network (WNN) to deal with both morphological and texture features, reaching up to 98% for accuracy.

In addition to the ensemble network derived from pre-trained models, the multimodal integration of diverse inputs significantly enhances the analysis of medical images from multiple perspectives. An Auto-weighting Multimodal (AWMM) framework is developed by Wang et al. [45], which utilizes four types of ultrasonography: B-mode, Doppler, Shear Wave Elastography (SWE), and Strain Elastography (SE). As a result, the AWMM framework achieves a classification accuracy of 95.43% on a dataset containing 1,616 multimodal image sets. Turning to the case of mammograms, Song et al. [46] introduced a multiview multimodal network (MVMM-Net) for breast cancer diagnosis. Due to the limitations of traditional single-view models, their framework aims to improve on other multimodal approaches by simultaneously processing images from multiple views and modalities. The experiments show an impressive result of 96.59% in terms of accuracy, with equally high sensitivity and specificity.

Furthermore, combining numerous imaging techniques may advance diagnostic performance, which allows for the learning of various types of features from diverse perspectives. A semi-automated multimodal classification system integrating mammogram and ultrasound imaging was developed by researchers [47] to enhance breast cancer detection. Through combining forty-two grayscale features extracted from both modalities, their innovative system substantially enhanced the ability to distinguish benign from malignant tumors. The experiments are conducted on a new joint dataset, which demonstrates a remarkable classification accuracy of 98.84% by using a cubic SVM. Atrey et al. [48] utilize CNN as an advanced extractor to process the features of ultrasound and mammograms. This brings an advantage in improving the accuracy and reliability of CAD. The findings of this research have shown superior performance compared to single-modality systems - reaching 99.35%.

Recently, graph convolutional neural networks (GCN) have highlighted the potential for improving the performance of image classification, especially for medical images. Indeed, the authors in [49] proposed a multi-graph convolutional neural network for breast cancer classification using mammograms. This network utilizes a blend of self-supervised and semi-supervised learning methods to improve feature representation. The results demonstrate that with only 40% to 60% of the data labeled, their models achieve classification accuracy with Area under the ROC Curve (AUC) scores of 0.97 and 0.98. Duong et al. [50] explore a method combining image processing and graph neural networks (GNNs) to enhance mammogram classification for breast cancer detection, tailored explicitly for Vietnamese women with denser breast tissue. Thanks to the utilization of conventional edge detection algorithms as a preprocessing, the adaptable extraction of unique characteristics from each mammogram is conducted. As a

TABLE 1. Summary of related works on breast cancer classification.

Author(s)	Dataset	Methodology	Result
Miron et al. [38] - 2022	BUSI (Ultrasound) - 2 classes	Feature extraction from ROIs and multi-layer network for classification	Validation accuracy: 86%
Yu et al. [39] - 2021	Self-collected (Ultrasound) - 2 classes	Pretrained Inception-V3 model	Accuracy: 85%, AUC:87.2%
Byra [40] - 2021	BUSI (Ultrasound) - 2 classes	Pretrained ResNet101 model	Accuracy: 95.43%, Sensitive: 96.15%, Specificity: 94.77%, Precision: 94.34%, F1-score: 95.24%
Ayana et al [41] - 2022	Mendeley dataset (Ultrasound) - 2 classes; MT-Small dataset (Ultrasound) - 2 classes	Mutistage models - EfficientNetB2, InceptionV3, and ResNet50	For Mendeley, test accuracy: 99 % For MT-Small, test accuracy: 98.7 %
Zhuang et al. [42] - 2021	BUSI (Ultrasound) - 2 classes	VGG11, VGG16, VGG19, ResNet101, DensNet121, and DensNet161	Accuracy: 95.48%, Sensitive: 93.92%, Specificity: 98.33%, Precision: 98.11%, F1-score: 95.71%
Khanna et al. [43] - 2021	BUSI (Ultrasound) - 2 classes	ResNet50 and BGWO are used for feature extraction, SVM is classifier	Accuracy: 84.9%, AUC: 0.97
Ahila et al. [44] - 2022	BUSI (Ultrasound) - 2 classes	GWO and WNN are applied for feature selection and classification	Accuracy: 98%, Sensitive: 98.8%, Specificity: 95.9%
Wang et al. [45] - 2020	Self-collected (Ultrasound) - 2 classes	Combining 4 types of ultrasonography (B-mode, Doppler, SWE, SE) and tuning by RL	Accuracy: 95.43%, Sensitive: 96.15%, Specificity: 94.77%, Precision: 94.34%, F1-score: 95.24%
Song et al. [46] - 2021	Self-collected (Mammogram) - 2 classes	Multimodal features from CC and MLO view are extracted. Res2Net50 is used as backbone for framework	Accuracy: 96.59%, Sensitive: 96.39%, Specificity: 96.35%, Precision: 96.83%, F1-score: 0.966
Atrey et al. [47] - 2023	Self-collected (Mammogram and Ultrasound) - 2 classes	Multimodal classification using the feature fusion from mammogram and ultrasound. SVM is employed in classification phase	Accuracy: 98.84%, Sensitive: 98.15%, Specificity: 99.33%, AUC: 99%
Atrey et al. [48] - 2023	Self-collected (Mammogram and Ultrasound) - 2 classes	The features of ultrasound and mammogram are extracted by CNN and the applied classifier is LSTM	Accuracy: 99.35%, AUC: 99%
Ibrahim et al. [49] - 2023	CBIS-DDSM (Mammogram) - 5 classes	Patches of images are converted to graph data, which are then classified by GNN.	AUC accuracy: 0.98
Duong et al. [50] - 2023	Self-collected (Mammogram) - BIRAD: 8 classes - Breast density type: 4 classes	Edge detection algorithm is used to perform graph transformation. GNN is then applied to perform classification	For BIRAD, Accuracy: 100% For Breast density type, Accuracy: 100%
Chowa et al. [51] - 2024	BUSI (Ultrasound) - 2 classes	Ten handcrafted features are extracted and classified by GNN model	Peak test accuracy: 99.48% Precision: 100%, Recall: 100%

result, GNNs can classify absolutely the BI-RADS scores and breast density types. For dealing with ultrasound images, Chowa et al. [51] developed a graph neural network optimized for breast cancer diagnosis. Their methodology concentrates on extracting and integrating ten clinically essential features into a graph dataset for modeling with GNN. Besides, features such as solidity, GLCM entropy, and circularity are considered vital in differentiating benign from malignant tumors. By defining relationships between image nodes using Spearman correlation coefficients, the model achieves high accuracy with a correlation threshold of 0.95.

III. PROPOSED APPROACH

In this work, an advanced approach for the classification of breast cancer through the use of ultrasound images by employing a node-based GCN is discussed. The proposed node-based graph modeling process for breast ultrasound is illustrated in Figure 1. Initially, ultrasound images are segmented into superpixels using the SLIC algorithm, which clusters pixels based on color similarity and spatial proximity, thereby simplifying the computational process while preserving essential image features. These superpixels are considered nodes in a graph where edges are established

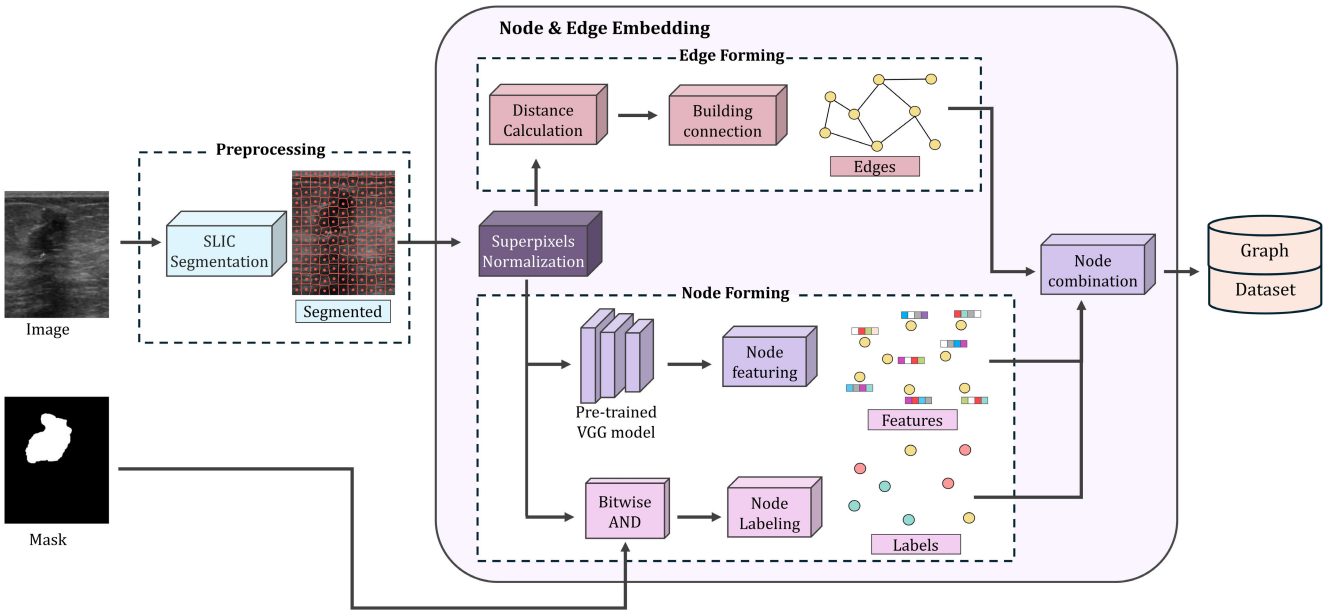


FIGURE 1. The proposed node-based graph modeling process for breast ultrasound image.

based on the minimum k-nearest neighbors by intensity and spatial distances. In order to provide more information for graph data, features for each node are extracted by the superpixel region to a 32 × 32 pixel frame and processed through a pre-trained VGG model. This process generates a 512-dimensional feature vector for each node. Besides, nodes are labeled as normal, benign, or malignant based on their overlap with annotated regions in corresponding mask images. To perform the node-level classification, the graph-structured data is then input into a GCN. The architecture consists of multiple graph convolution layers and concludes with a softmax layer, which allows the model to learn and generalize from the complex interrelationships among nodes. Finally, image-level classification is achieved by aggregating these classified nodes by a voting rule, providing a comprehensive identification of the ultrasound images for breast cancer classification.

A. DATASETS

This research utilizes two datasets to identify breast cancer by ultrasound images. The first dataset is the Breast Ultrasound Images Dataset (BUSI), which was gathered by Al-Dhabyani et al. [52]. This comprises 780 images of female patients aged between 25 and 75 years. The images are classified into three categories: normal (133 images), benign (437 images), and malignant (210 images). Besides, ground truth masks were provided for both the benign and malignant categories to highlight the abnormal regions of the images. On the other hand, the Breast Ultrasound Image dataset from Brazil (BUSBRA) collected by Gomez-Flores et al. [53] is considered as the second dataset for our study, which consists of 1,875 images. Unlike BUSI, BUSBRA categorizes images into two main groups: benign (1,268 images) and malignant (607 images). Each image in this dataset includes a binary ground truth mask that precisely highlights tumor locations. Thus, our study aims to classify the ultrasound image in the categories of benign, malignant, and normal in the BUSI dataset, while only benign and malignant are examined in the BUSBRA dataset. The selection of BUSI dataset is based on the widespread use in prior research, which allows for easy comparisons with existing studies. In contrast, BUSBRA is considered as a relatively new dataset, which is able to provide fresh insights and a huge amount of number samples. Figure 2 illustrates the representative samples from these datasets.

B. PREPROCESSING

Simple Linear Iterative Clustering (SLIC) is proposed by Achanta et al. [54] as an algorithm applied for segmenting images based on superpixel. It splits an image into several segments by grouping pixels together according to their related characteristics, such as color or intensity. This provides a computational efficiency benefit by decreasing the amount of distance calculations, which is required in comparison to conventional clustering algorithms. In this work, SLIC algorithm is utilized to divide the breast ultrasound image into smaller sections called superpixels, employing the properties of individual pixels. This process comprises three primary steps:

1) INITIALIZATION

The initial step involves taking an image consisting of N pixels, which is divided into a regular grid of small patches. The dimensions of these patches are determined according to

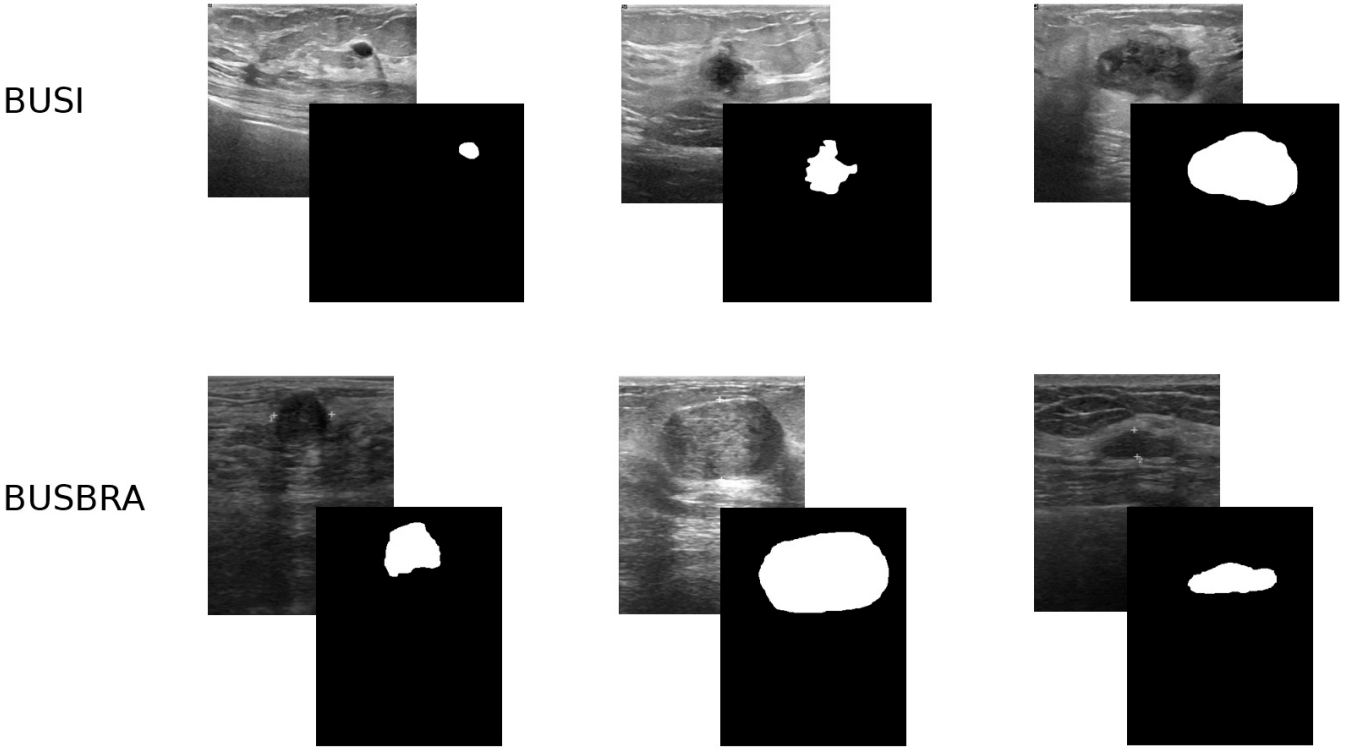


FIGURE 2. Sample breast ultrasound images and their ground truths from two datasets.

the specified number of superpixels K. Therefore, the formula for the size of the grid interval is given by (1).

$$S = \sqrt{\frac{N}{K}} \tag{1}$$

This initializes the center $C_k = [l_k, a_k, b_k, x_k, y_k]$ for each superpixel, where l, a, b are the CIELAB color space and x, y are the pixel coordinates. In order to improve adherence to borders, the centers are positioned at the lowest gradient position within a 3×3 neighborhood to prevent placing on an edge and minimize the possibility of superpixel seeding with a noisy pixel.

2) k-MEANS CLUSTERING

Each pixel in the images is connected to the adjacent cluster center during the assignment step. SLIC takes into account both similarity and spatial proximity to guarantee compact, nearly uniform superpixels. The distance metric D is a combination of color and spatial distance, which allows SLIC to balance color similarity and spatial closeness. The distance metric is defined in (2).

$$D = \sqrt{d_{color}^2 + \left(\frac{d_{space}}{S}\right)^2 \cdot m^2} \tag{2}$$

where d_{color} is the Euclidean distance in color space and d_{space} is the Euclidean distance in the image plane. The formulas for d_{color} and d_{space} are shown in (3) and (4), respectively, m is a compactness parameter controlling the ratio between two distances.

$$d_{color} = \sqrt{(l - l_c)^2 + (a - a_c)^2 + (b - b_c)^2} \tag{3}$$

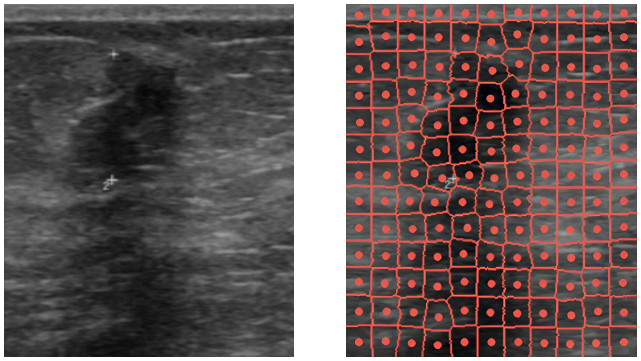
$$d_{space} = \sqrt{(x - x_c)^2 + (y - y_c)^2} \tag{4}$$

where l, a, b and l_c, a_c, b_c are the CIELAB color values of the pixel and cluster center, while x, y and x_c, y_c are the coordinates of the pixel and cluster center. Once all the pixels are matched with their cluster center, the next task is to update the position and color of each cluster center by the current assignments. This involves recalculating each cluster center as the mean vector of all pixels belonging to the cluster. As a result, this process boosts the accuracy of superpixel clusters to achieve better image alignment. These assignment and update steps are iterated until a point of convergence is reached. Indeed, the convergence is determined as a slight change in the cluster center positions between consecutive iterations.

3) POST-PROCESSING

Due to varying image features and noise, superpixels might be spatially disconnected. By ensuring that every superpixel is interconnected and forms a single and contiguous region, the connectivity check is then performed after the initial clustering. This can be done by using graph traversal techniques on superpixel label for identifying connected

components. In the case of appearing the disconnected pixels, reassignment is processed to the nearest neighboring superpixel based on spatial proximity. This operation might require several iterations in order to ensure no further disconnected components in the image. Thanks to these operations, the problem of disconnected components is resolved, which brings the increase coherence and usability in segmentation. Thus, each superpixel adherently reflects a continuous part of the image, and the overall result from these superpixels is more reliable for the next stage of processing As a result, the superpixels are obtained, and the centroid of each region is computed in preparation for the subsequent steps. Figure 3 shows the illustration of applying SLIC.



Original image After applying SLIC

FIGURE 3. Visualization of SLIC process.

C. NODE AND EDGE REPRESENTATIONS

Prepared graph data is required before using GCN, which involves defining the nodes, edges, and features (attributes) associated with them. A graph $\mathcal{G}$ is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{v_1, v_2, \ldots, v_N\}$ denotes as a set containing N vertices (nodes) and $\mathcal{E} = \{e_1, e_2, \ldots, e_M\}$ implies as a set having M edges. Each single node $v \in \mathcal{V}$ has a feature vector x_v, and the features of entire graph data are represented in a feature matrix $X \in \mathbb{R}^{N \times F}$, where N and F are the number of nodes and features per node, respectively. To ensure consistent processing across a range of stages, every superpixels is resized to a constant 32×32 pixels in size. This resizing stage is crucial in order to standardize the input for further processing, which ensures that each superpixel region can be analyzed consistently. The following procedures are separated into two paths. The first path is to form node edges by transforming the two-dimensional pixel matrix form of superpixels to vectors, while the other path preserves the original dimension for forming node features.

Initially, the nodes in the graphical representation are constructed using these superpixels, and the edges are produced by considering both the pixel intensities and the spatial positions of the superpixels. To process this, these resized superpixels are flattened into vectors by concatenating the pixel values of each row in the region. As a result, this transforms the two-dimensional pixel matrix into a one-dimensional vector. Thus, each superpixel region is systematically converted into a vector representation $S = \{s_1, s_2, \ldots, s_d\}$, where each element represents an individual pixel from the superpixel and d is determined by the number of pixels in the resized superpixel. The value of d is fixed for all superpixels due to the previous resized process, resulting in all vectors having the same length. The calculation of the difference in pixel intensities between two nodes, v_a and v_b, is determined by equation (5).

$$d_{intensity}(v_a, v_b) = \sqrt{\sum_{i=1}^{d} (s_{a_i} - s_{b_i})^2} \tag{5}$$

Similarly, the spatial connection between two superpixels is measured using the location distance formula (6).

$$d_{location}(v_a, v_b) = \sqrt{(x_a - x_b)^2 + (y_a - y_b)^2} \tag{6}$$

where x_a, x_b, y_a, and y_b are the centroid coordinates of any pair of pixels, providing a measure of physical proximity in the image space. The graph is constructed by establishing edges between each node and its top K nearest neighbors based on these distances, which are normalized and aggregated according to the summation of these two distances to balance the influence of intensity similarity and spatial proximity, as described in (7).

$$d_{total}(v_a, v_b) = \frac{d'_{intensity} + d'_{location}}{2} \tag{7}$$

where $d'_{intensity}$ and $d'_{location}$ are the normalized distance in the range of 0 and 1. The graphical illustration for edge forming is depicted in Figure 4.

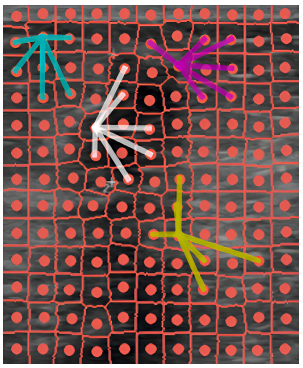


FIGURE 4. Visualization of forming edges.

The relationships between nodes are shown by an adjacency matrix $\mathcal{A}$, which $\mathcal{A}_{ij} = 1$ in case of existing an edge from node i to node j, and $\mathcal{A}_{ij} = 0$ otherwise. Normally, the adjacency matrix is normalized by symmetric normalization to facilitate the learning process. This process is calculated by (8).

$$\hat{\mathcal{A}} = \tilde{\mathcal{D}}^{-\frac{1}{2}} \tilde{\mathcal{A}} \tilde{\mathcal{D}}^{-\frac{1}{2}} \tag{8}$$

where $\hat{\mathcal{A}}$ is the adjacency matrix of graph $\mathcal{A}$ with added self-connection I_N and $\tilde{\mathcal{D}}$ is the degree matrix of $\tilde{\mathcal{A}}$, which are formulated in (9) and (10), respectively.

$$\tilde{\mathcal{A}} = \mathcal{A} + I_N \tag{9}$$

$$\tilde{\mathcal{D}}_{ii} = \sum_j \tilde{\mathcal{A}}_{ij} \tag{10}$$

On the other hand, in order to effectively embed the node features, each superpixel region is subjected individually to further analysis using a pre-trained VGG19 model [16], which has been trained on a comprehensive image dataset, allowing the extraction of high-level features from the image data. Thanks to the process of resizing to 32×32 pixels in previous steps, the current superpixels are compatible with the VGG model, since this model requires fixed-size input. As a result, the output for each superpixels is formed as a 512-dimensional vector, which is treated as the node feature for the corresponding node. This advanced feature extraction leverages the framework to enhance the understanding of the local context from the short-range pixels.

Furthermore, node labeling is a crucial step in preparing the graph representation by utilizing the mask image, which is prepared alongside the original ultrasound image, delineating regions classified as benign, malignant, or normal tissue. These regions are predefined by a radiologist or derived from diagnostic findings associated with the ultrasound images from the dataset. The labels of the nodes are assigned based on the congruence of their central positions (x_c, y_c) with the annotated areas of interest within the corresponding mask image. If the coordinates (x_c, y_c) reside within a region denoted as benign on the mask image, the node receives a label of 0. Conversely, if (x_c, y_c) is situated within a region identified as malignant, the node is assigned a label of 1. Alternatively, the node is labeled as 2, which is categorized as normal in case of falling in a region not marked by any specific condition (i.e., outside benign or malignant areas). Thanks to the segmented results from the SLIC, the superpixels receive clearer delineation between tumor and non-tumor areas, which might benefit in enhancing the accuracy of region boundary. Besides, the employment of centroids for labeling also can bring more representative of the superpixel's overall characteristics. The node labeling can be mathematically described by (11), while the visualization of the labeling process is illustrated in Figure 5.

$$L = \begin{cases} 0, & \text{if } (x_c, y_c) \in \text{Benign region}, \\ 1, & \text{if } (x_c, y_c) \in \text{Malignant region} \\ 2, & \text{otherwise.} \end{cases} \tag{11}$$

D. GRAPH CONVOLUTIONAL NETWORK

Graph Convolutional Network (GCN) is introduced as an advanced version of neural networks which is designed to process the graph-structured data. In contrast to conventional deep learning approaches, which are mainly designed to work with Euclidean data (images, text sequences), this type of networks is able to thrive on graph-structured data encapsulating complex relationships and interdependencies.

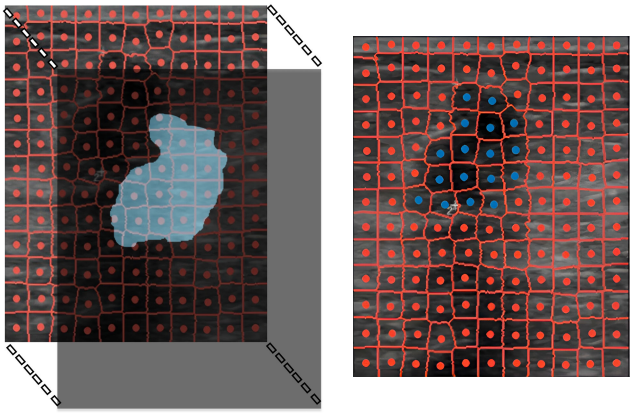


FIGURE 5. Visualization of the labeling process.

In order to implement GCNs for node and edge representation, the adjacency matrix of the graph and node features are provided as inputs. The neighbor information for each node is subsequently aggregated by the GCN through a sequence of convolutional layers. By aggregating the representations of its neighbouring nodes, each layer updates the representation of a node, thereby effectively capturing both local and global graph structures. In contrast to conventional CNNs, the convolution operation in GCNs is executed by employing a mixed method consisting of linear transformations and non-linear activations which traverse the nodes of the graph. Consider the primary case of GNN, the forward pass of a basic ℓ-th layer can be described mathematically in (12).

$$\begin{aligned} \tilde{h}_i^{(\ell+1)} &= \text{AGGREGATE}\left(\left\{h_j^{(\ell)} : j \in \mathcal{N}(i)\right\}\right), \\ h_i^{(\ell+1)} &= \text{UPDATE}\left(\tilde{h}_i^{(\ell+1)}\right) \end{aligned} \tag{12}$$

where $h_i^{(\ell)}$ is the feature representation of node i at layer ℓ, while AGGREGATE and UPDATE are separately defined by the purpose of the network. $\mathcal{N}(i)$ denotes the neighbor of i, and the initial node features $h_i^{(0)}$ are often set to the raw features of the nodes x_i. In the case of GCN, the operation can learn node representations employing a non-linear transformation and an aggregator with a fixed weight. Thus, comprising (12) in the matrix form is possible for GCN, defined in (13).

$$h_i^{(\ell+1)} = \sigma(\hat{\mathcal{A}} h^{(\ell)} \mathcal{W}^{(\ell)}) \tag{13}$$

where $\mathcal{W}^{(\ell)}$ represents a learnable weight matrix at layer ℓ, and σ is a non-linear activation function.

The proposed GCN architecture in this work is shown in Figure 6. To begin with, the adjacency matrix and the corresponding node feature vectors of graph data are processed before feeding into three GCN layers. Each layer transforms the input features to capture more complex relationships between the nodes. The initial GCN layer with the RELU activation function transforms the input node

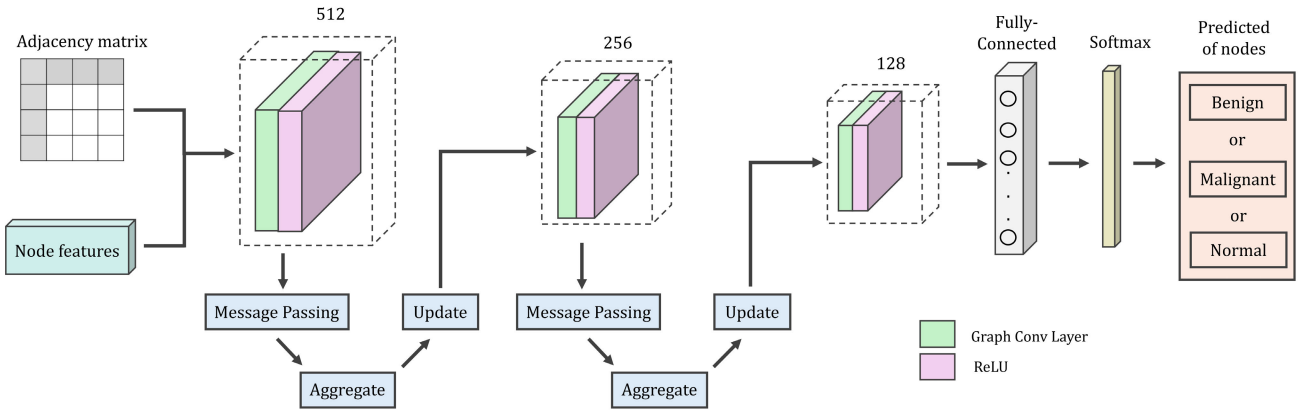


FIGURE 6. The proposed graph convolutional neural network model for node-level classification.

features $h^{(0)} = h$ into a new set of feature $h^{(1)}$ which is described in (14).

$$h^{(1)} = \sigma(\hat{A}h^{(0)}\mathcal{W}^{(1)}) \tag{14}$$

This transformation involves a matrix multiplication of the node features by a learnable weight matrix $\mathcal{W}^{(1)}$ and an aggregation of neighboring node features via multiplication by the normalized adjacency matrix. The process is repeated for the second and third graph convolutional layers, each with their own weight matrices $\mathcal{W}^{(2)}$ and $\mathcal{W}^{(3)}$ as formulated in (15) and (16).

$$h^{(2)} = \sigma(\hat{A}h^{(1)}\mathcal{W}^{(2)}) \tag{15}$$

$$h^{(3)} = \sigma(\hat{A}h^{(2)}\mathcal{W}^{(3)}) \tag{16}$$

After traversing three GCN layers, each node in the network possesses an enhanced feature vector that encapsulates the combined information from its adjacent nodes. Afterward, the subsequent stage involves transmitting the learned node embeddings through a fully connected layer, which performs a linear transformation to connect all the neurons in previous layers, acting as a combination of features. To perform classification, the softmax layer converts the output of the fully connected layer into probability distribution output for multi-class data, which is defined in (17).

$$P_{ij} = \frac{e^{h_{ij}^{(fc)}}}{\sum_k h_{ik}^{(fc)}} \tag{17}$$

where P_{ij} is the predicted probability of node i belonging to class j, and k represents the index of a class.

E. IMAGE-LEVEL CLASSIFICATION

In the graph-based representation of ultrasound images, nodes are derived from superpixels generated through image segmentation. However, obtaining accurate and reliable image-level classifications from comprehensive node-level analyses is crucial in the context of medical imaging for the purpose of diagnosing breast cancer via ultrasound.

To obtain a comprehensive understanding of the distribution of various nodes in ultrasound images, histograms are generated for benign and malignant nodes using a subset of the dataset, as illustrated in Figure 7 and Figure 8. Notably, benign nodes tend to be fewer in number, typically one or two per image, reflecting the smaller size of benign tumors. In contrast, malignant nodes are more numerous, frequently exceeding two, which indicates the larger and more invasive characteristics of malignant tumors.

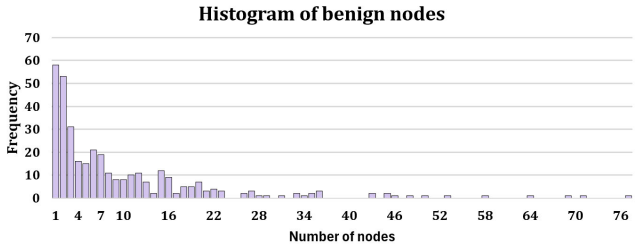


FIGURE 7. The statistics of benign nodes.

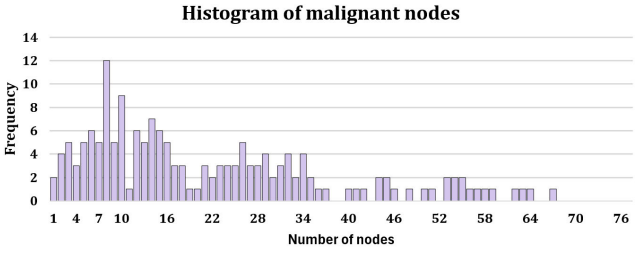


FIGURE 8. The statistics of malignant nodes.

The process of deriving a holistic image-level classification from discrete node-level data involves two critical steps:

1) NODE COUNT AGGREGATION

Each node classified as benign, malignant, or normal contributes to the cumulative count for each category, which is denoted as c_0, c_1 and c_2, respectively. These counts are

crucial indicators of the extent and nature of the pathology present in the image.

2) IMAGE-LEVEL CLASSIFICATION RULE

The classification of the entire image is determined based on the relative counts of benign and malignant nodes using a decision-making algorithm. The flowchart of this algorithm is shown in Figure 9. This algorithm prioritizes the classification of an image as benign if the count of benign nodes (c_0) is greater than or equal to the count of malignant nodes (c_1). This decision reflects a major voting rule that the label of image level is mainly based on the presence of a large number of corresponding nodes. On the other hand, in case that malignant nodes are more prevalent ($c_1 > c_0$) and their count is above a threshold of two, the image is classified as malignant, aligning with observations that malignant tumors tend to be larger and involve more extensive areas of the tissue. In the absence of both conditions being satisfied or the detection of any pathological nodes, the image is classified as normal.

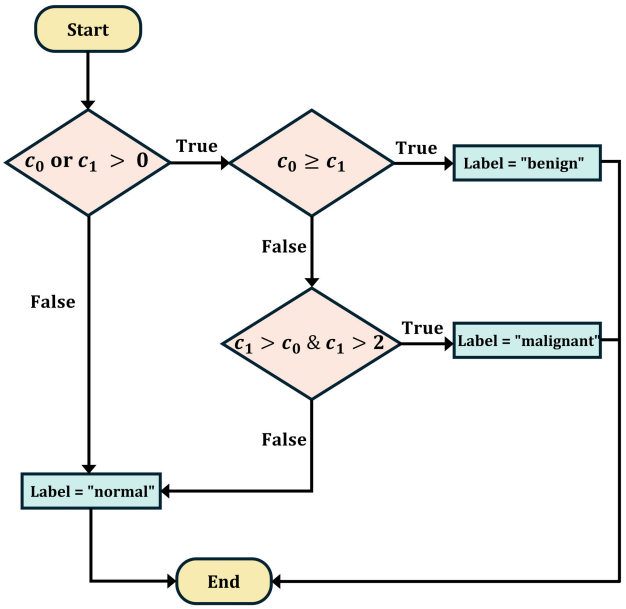


FIGURE 9. The flowchart of image-level classification.

IV. RESULTS AND DISCUSSION

A. EXPERIMENTAL SETUP

In order to assess the efficacy of the suggested methodology, experiments are carried out on a workstation featuring an Intel Core i7-12700 CPU, 128 GB of RAM, and an NVIDIA RTX 3090 GPU with 24 GB of VRAM. The conducted environment is established on Python 3.10 with Pytorch version 2.0.1 and PyTorch Geometric version 2.4.0. This configuration provided sufficient computational resources to conduct evaluation experiments efficiently. The proposed approach is implemented across two distinct breast ultrasound image datasets, which are partitioned into training and testing sets using an 80/20 split ratio. This division ensures sufficient data for training while reserving a separate portion for evaluating unseen data. To ensure that the evaluation process is conducted in an impartial and truthful way, the same partitioning of the dataset is rigorously applied from the node-level to image-level classifications. This guarantees that the integrity of the test data is not compromised during the transition between evaluation stages. The distribution summary of each class for datasets is presented in Table 2. While the number of images in the "normal" class is the smallest for the BUSI dataset, they comprise the greatest quantity in terms of nodes. This observation can be attributed to the fact that the area occupied by abnormal regions tends to be smaller compared to normal areas. As a consequence, the number of nodes classified as "benign" and "malignant" is comparatively lower than that of the "normal" class. Regarding the BUSBRA dataset, although comprising only two classes at the image level, namely "benign" and "malignant", at the node-based level, the consideration of background nodes as normal introduces the presence of the "normal" class in this level as well.

TABLE 2. Summary of class distribution of two datasets.

Class	BUSI Dataset		BUSBRA Dataset	
	Number of images	Number of nodes	Number of images	Number of nodes
Benign	437	7,735	1,268	12,906
Malignant	210	4,516	607	10,876
Normal	133	108,310	-	256,738

During the training phase, the model was optimized using the Focal Loss function to address class imbalance by down-weighting well-classified examples. Mathematically, the Focal Loss is defined in (18).

$$FL(p_t) = -\alpha_t(1 - p_t)^{\gamma} \log(p_t) \tag{18}$$

where p_t is the estimated probability of model for each class, and α_t is a weighting factor for each class to balance the importance of the class. The formula for α_t for each class is defined by the ratio total samples S_{total} and the sample of each class S_{class}, which is calculated in the (19). The parameter γ is used to reduce the relative loss for well-classified examples, enabling the model to focus more on hard-to-classified examples.

$$\alpha_t = \frac{S_{total}}{S_{class}} \tag{19}$$

The applied optimizer for training is based on the Adam algorithm, which is known as the combination of Momentum and RMSprop. Thanks to the advantages of these methods, the process provided by Adam is effortless to reach the global minimum point without time-consuming in fluctuations across the local minimum points. Thus, the training process was conducted over 200 epochs with an initial learning rate set to 1×10^{-4}, meanwhile the trainable parameters for GCN

model are 295,683. The number of batch sizes is configured to adapt the VRAM of the given GPU.

The evaluation processes are further focused on key performance metrics for classification results. Let $L = (L_1, L_2, \ldots, L_n)$ represents the set of actual labels, and $C = (C_1, C_2, \ldots, C_n)$ is the set of predicted labels for the dataset with n instances. Accuracy measures the overall correctness of the model and is calculated as the ratio of correct predictions to the total number of predictions, as mentioned in (20).

$$\text{Accuracy} = \frac{\sum_{i=1}^{n} \mathbf{1}(L_i = C_i)}{n} \tag{20}$$

where $1(L_i = C_i)$ is an indicator function that returns 1 if the predicted label C_i is equal to the true label L_i, and 0 otherwise. Precision is the ratio of correctly predicted positive instances to the total predicted positives of the dataset, while recall evaluates the ability of a model to correctly identify all relevant instances. The formulas for these metrics are shown in (21) and (22), respectively.

$$\text{Precision} = \frac{1}{k} \sum_{j=1}^{k} \frac{TP_m}{TP_m + FP_m} \tag{21}$$

$$\text{Recall} = \frac{1}{k} \sum_{j=1}^{k} \frac{TP_m}{TP_m + FN_m} \tag{22}$$

where TP_m is the number of true positives (correctly predicted as class m), and k is the number of classes. FP_m is the number of false positive results, whereas FN_m denotes the quantity of false negative results. Thus, the F-Score provides a useful compromise between recall and precision, particularly when the distribution of classes is irregular. This metric is the harmonic mean of the two mentioned metrics, as described in (23).

$$\text{F-score} = \frac{1}{k} \sum_{j=1}^{k} 2 \times \frac{\text{Precision}_j \times \text{Recall}_j}{\text{Precision}_j + \text{Recall}_j} \tag{23}$$

B. QUANTITATIVE ANALYSIS OF NODE-LEVEL CLASSIFICATION

The performance metrics for node-level classification are employed to evaluate the strength of node based on the proposed framework. This analysis offers an in-depth understanding of the potential uses of nodes, thus providing a robust foundation for advancing to further procedures. To gain additional insight into the impact of these interactions, it is critical to examine the effects of manipulating the quantity of connections between nodes on the performance of model. Our experiments are conducted with different k connections between each node to evaluate the essential of interconnection in a graph. Table 3 and Table 4 present the summary performance of the BUSI and BUSBRA datasets, respectively.

Overall, the total number of connections for each dataset starts at a high value, over 200,000 for the BUSI dataset and 500,000 for the BUSBRA dataset. These figures expanded, reaching approximately 1 million connections in the BUSI dataset and surpassing 2 million in the BUSBRA dataset at $k = 8$. This rise in connections necessitates a corresponding increase in computational resources to manage the heavy load of processing. In the case of BUSI dataset, the accuracy metric exhibits an impressive boost from 98.63% at $k = 2$ to an optimal of 99.80% by $k = 6$ and remains constant afterwards. The appearance of this plateau suggests that there is a saturation point in accuracy, which additional connections do not translate into further improvements. Meanwhile, a significant enhancement from 88.81% at $k = 2$ to 97.79% at $k = 6$ and $k = 8$ is presented in precision, which demonstrates a substantial reduction in false positives with increased connectivity. In contrast, recall shows minimal improvement but maintains exceptionally high value that consistently exceeds 99% owing to the dominance of samples from the ''normal'' class. The F-score also sees a remarkable rise from 93.70% to 98.54%, indicating an overall advancement in the efficiency and reliability of model as the growth in number of connections. On the other hand, the BUSBRA dataset starts with a higher baseline for most performance metrics. Early optimization in terms of true positive and true negative identifications is indicated by the fact that accuracy rapidly saturates at 99.90% from $k = 4$ onwards. Precision exhibits a sharp and significant increase, reaching a peak of 99.41% at $k = 4$, which is maintained through $k = 8$. Similar to BUSI, recall is nearly perfect for all values of k, exceeding 99.5%. Furthermore, the F-Score increases from 97.25% to 99.56%, indicating that the model possesses superior predictive capability as k increases.

TABLE 3. The performance metrics of BUSI dataset for node-level classification.

	Number of k connections			
	$k = 2$	$k = 4$	$k = 6$	$k = 8$
Total number of edges	241,122	482,244	723,366	964,488
Accuracy	98.63%	99.26%	99.80%	99.80%
Precision	88.81%	92.83%	97.79%	97.79%
Recall	99.17%	99.17%	99.29%	99.29%
F- score	93.70%	95.90%	98.54%	98.54%

TABLE 4. The performance metrics of BUSBRA dataset for node-level classification.

	Number of k connections			
	$k = 2$	$k = 4$	$k = 6$	$k = 8$
Total number of edges	561,040	1,122,080	1,683,120	2,244,160
Accuracy	99.11%	99.90%	99.90%	99.90%
Precision	94.93%	99.41%	99.41%	99.41%
Recall	99.68%	99.72%	99.72%	99.72%
F- score	97.25%	99.56%	99.56%	99.56%

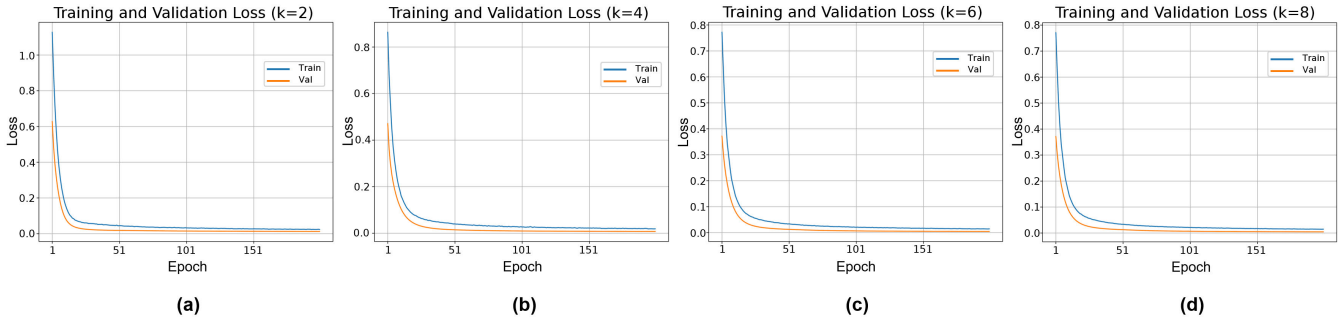


FIGURE 10. Training and validation loss of BUSI dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

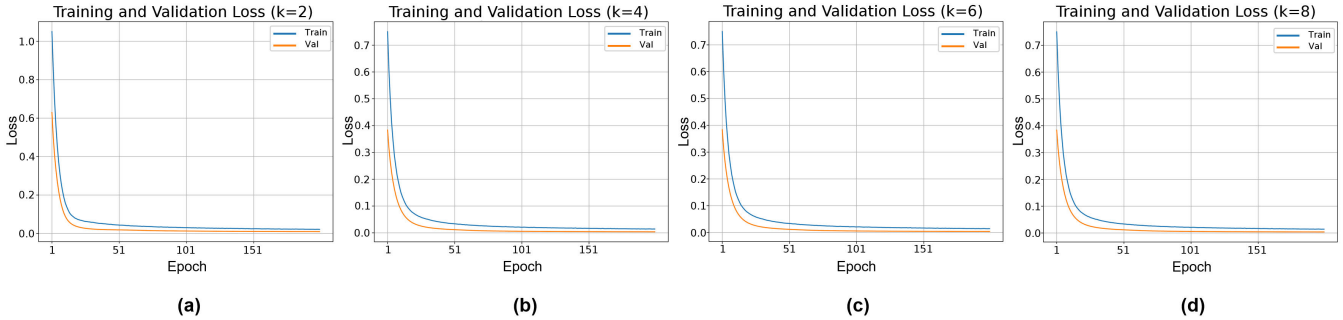


FIGURE 11. Training and validation loss of BUSBRA dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

Furthermore, the training and validation loss curves for the BUSI and BUSBRA datasets, depicted in Figure 10 and Figure 11. In both datasets, the plots demonstrate a notable decrease in both training and validation loss as the number of epochs increases in all the cases of k. It is observed that the convergence of training and validation loss occurs more rapidly and with a tighter convergence gap, which indicates the reduction of overfitting as k increases. Specifically, the curves denoting $k = 6$ and $k = 8$ exhibit a substantial degree of flattening after around 50 epochs. This observation implies that the model attains an optimal learning state earlier in case of the inclusion of more connections between nodes. The detailed evaluation of the confusion matrices for the BUSI and BUSBRA datasets as presented in Figure 12 and Figure 13, respectively. In the case of BUSI dataset, a consistent improvement in true positive rates for Benign (B) and Normal (N) classifications is observed as the connectivity parameter k increases. In particular, the accuracy for benign classifications remains notably high across all k values, reaching 99.68% for $k = 6$ and 8. Conversely, the misclassifications experience a significant decline, as evidenced by the diminished occurrences of false positives and false negatives in the matrices. In contrast, the BUSBRA dataset shows pronounced stability in classification accuracy across different k settings for benign and normal nodes, maintaining an accuracy exceeding 99% in nearly all instances. The indexes of classifying true Benign (B) and Normal (N) instances show minimal fluctuation, maintaining a remarkable steadiness with a slight increase in false negatives as k increases.

These results underline the adaptive capability of node-based models to perform breast cancer classification. Besides, the performance metrics are improved by leveraging increased graph connectivity. This ability to enhance performance metrics through the strategic increase in k connections indicates that the network is efficiently utilizing the additional edges to capture more complex and hidden patterns in the dataset, which are essential for accurate classification. However, the saturation of evaluation metrics when increasing to a certain number of connections implies the presence of a threshold effect where beyond a certain level of connectivity. Beyond this level, further connections do not improve outcomes but instead may lead to increased computational demands. Alternatively, the high recall across both datasets ensures that the model is reliably sensitive to detecting the majority of malignant cases, which is critical in breast cancer screening to avoid overlooking treatable cancers.

C. QUANTITATIVE ANALYSIS OF IMAGE-LEVEL CLASSIFICATION

In the context of evaluating the performance of the end-to-end process, the image-level classification is employed by the voting rule mentioned in the previous section. The analysis is conducted on the basis of the node-level stage, which focuses on the impact of varying the number of k connections on the

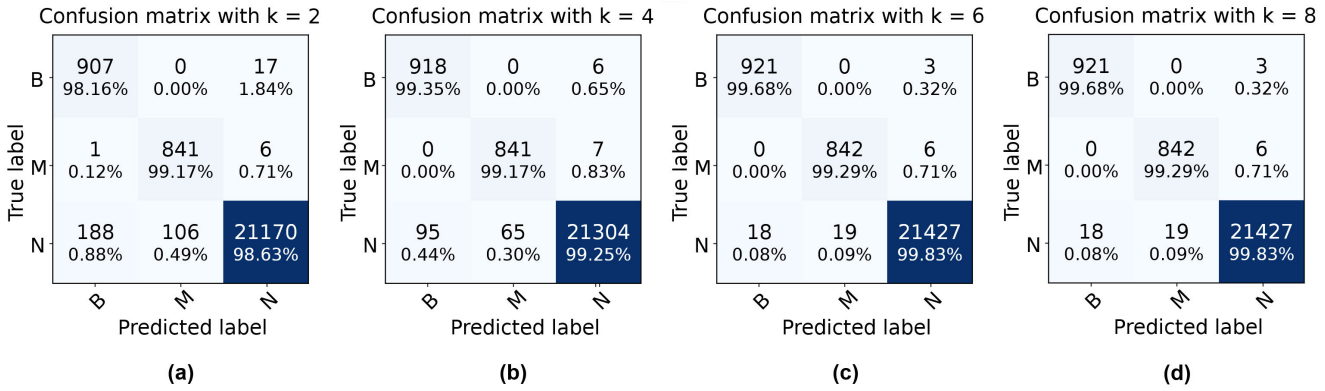


FIGURE 12. The confusion matrix of node-level classification for BUSI dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

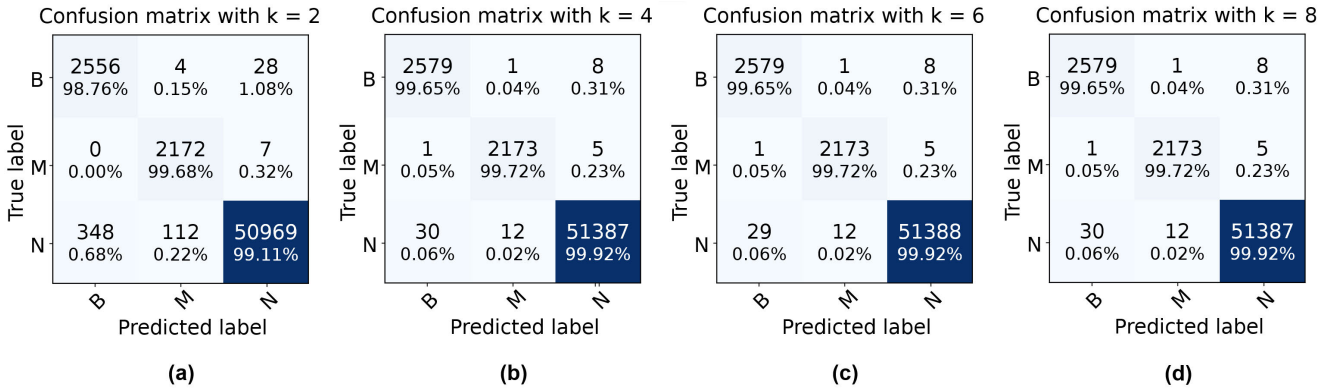


FIGURE 13. The confusion matrix of node-level classification for BUSBRA dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

overall effectiveness, as summarized in Table 5 and Table 6. Notably, the BUSI dataset demonstrates a clear trend of a remarkable improvement in accuracy metrics, with the value reaching 98.73% at $k = 6$ and $k = 8$. At the same time, the precision reaches perfect levels of 100% from $k = 4$ and remains at high levels during all subsequent k values, which indicates an impressive performance in identifying positive cases as the node connections are strengthened. Despite minor disruptions, recall and F-score remain robust at over 95% in all evaluated k values, proving the efficacy of the model in identifying relevant instances. Contrastingly, the BUSBRA dataset demonstrates a comparable but divergent trend of barely declining accuracy levels after an initial 98.67% accuracy at $k = 2$ and down to 98.40% from $k = 4$. Despite this, precision remains perfect throughout all k values, thus ensuring the ability of the model to detect the positive classes. However, recall and F-score metrics take a slight negative trend, with recall values falling nearly 1% as k values increase. This trend indicates that the sensitivity of the model is decreasing, leading to a negative effect on dealing with false alarms.

To be more specific, the performance analysis of each class is displayed in Figure 14 and Figure 15, corresponding to BUSI and BUSBRA datasets, respectively. For BUSI dataset, the confusion matrices reflect a consistent pattern of high accuracy in predicting Benign (B) cases across nearly all values of k, achieving a percentage of 100%. Meanwhile, in Malignant (M) also witnesses a noticeable result, which only one case in Malignant is failed to classify correctly for $k = 2, 6$ and 8. In addition, the Normal (N) label shows significant misclassification issues with only 40.74% accuracy at initial. This scenario improves dramatically as k increases, showing substantial enhancement in correctly classifying normal cases - reaching above 96%. This suggests a refined model sensitivity and specificity in distinguishing between normal and pathological states in case of increasing the connections between nodes. On the contrary, there are only two classes in BUSBRA for image-level classification due to the limitation of the initial dataset. Analogue to BUSI, this dataset maintains a consistent 100% accuracy rate in classifying Benign (B) cases across all k values, which demonstrates the robustness in classification without error. However, the classification of malignant cases shows slight variability, with about 5% of malignant cases being misclassified as normal across $k = 4$ to $k = 8$. These misclassifications may be explained that the initial voting algorithm for image-level classification is completely based on three classes, whereas BUSBRA only consists of

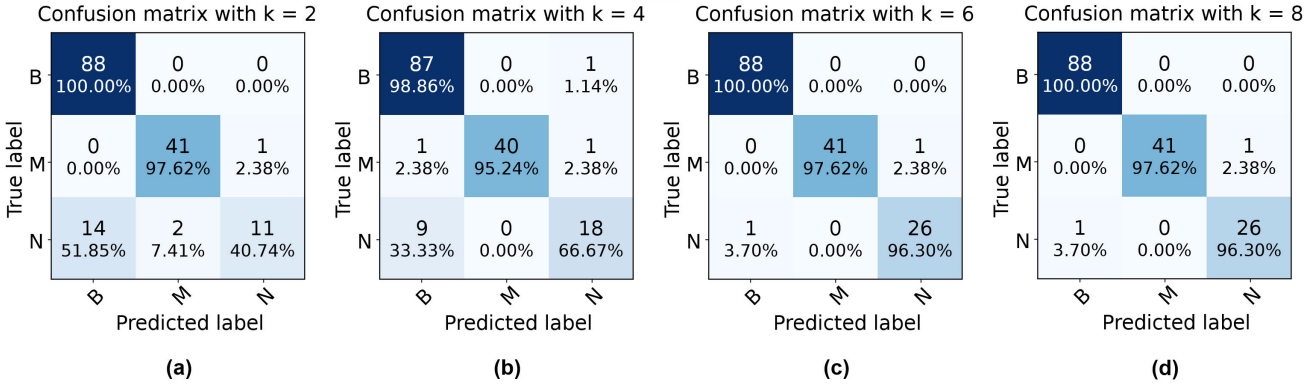


FIGURE 14. The confusion matrix of image-level classification for BUSI dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

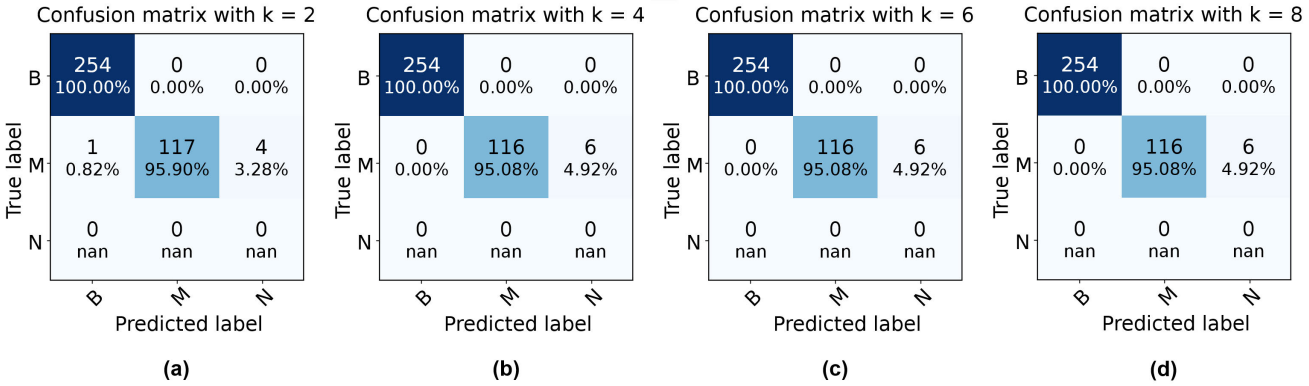


FIGURE 15. The confusion matrix of image-level classification for BUSBRA dataset with (a) $k = 2$, (b) $k = 4$, (c) $k = 6$, (d) $k = 8$.

TABLE 5. The performance metrics of BUSI dataset for image-level classification.

	Number of k connections			
	$k = 2$	$k = 4$	$k = 6$	$k = 8$
Accuracy	89.17%	92.36%	98.73%	98.73%
Precision	95.35%	100.00%	100.00%	100.00%
Recall	97.62%	95.24%	97.62%	97.62%
F- score	96.47%	97.56%	98.80%	98.80%

TABLE 6. The performance metrics of BUSBRA dataset for image-level classification.

	Number of k connections			
	$k = 2$	$k = 4$	$k = 6$	$k = 8$
Accuracy	98.67%	98.40%	98.40%	98.40%
Precision	100.00%	100.00%	100.00%	100.00%
Recall	95.90%	95.08%	95.08%	95.08%
F- score	97.91%	97.48%	97.48%	97.48%

Benign (B) and Malignant (M), leading to the existence of exceptional cases.

These results imply that although the increasing connectivity generally enhances the classification results, potential benefits are expected to become marginal after a certain threshold. Besides, it is evident that there have been no instances of Benign (B) cases being incorrectly identified as Malignant (M). In other words, the persistent ability of model is demonstrated to accurately classify benign cases as malignant while preventing the mistaken labeling of malignant cases as benign. This characteristic offers critical implications for clinical context since false positives of malignancy lead to unnecessary medical treatments. Consequently, the mentioned level of detailed examination aids in developing more robust and efficient diagnostic tools, which are essential for clinical practice and patient management.

D. ABLATION STUDIES

1) EFFECTIVENESS OF NUMBER OF CONNECTIONS

The analysis of various k connections has been described comprehensively in the previous parts, as depicted in Table 3 and Table 4 for both datasets in node-level. Besides, the image-level analysis is shown in Table 5 and Table 6. Overall,

TABLE 7. A comparison of node feature extraction by different pre-trained models.

	BUSI Dataset				BUSBRA Dataset			
	Accuracy	Precision	Recall	F- score	Accuracy	Precision	Recall	F- score
VGG19	**99.80%**	**97.79%**	**99.29%**	**98.54%**	**99.90%**	**99.41%**	**99.72%**	**99.56%**
ResNet50	97.78%	76.84%	99.76%	86.81%	98.01%	91.22%	99.13%	95.01%
MobileNetV2	83.05%	46.96%	97.52%	63.40%	73.19%	43.77%	96.65%	60.25%
DenseNet101	77.66%	27.13%	97.76%	42.47%	80.10%	30.76%	96.19%	46.61%

TABLE 8. A comparison of different parameters and options in training GCN.

	BUSI Dataset				BUSBRA Dataset			
	Accuracy	Precision	Recall	F- score	Accuracy	Precision	Recall	F- score
Activation function								
SELU	99.80%	97.23%	99.29%	98.25%	99.90%	99.36%	99.72%	99.54%
ReLU	**99.80%**	**97.79%**	**99.29%**	**98.54%**	**99.90%**	**99.41%**	**99.72%**	**99.56%**
LeakyRELU	99.79%	97.45%	99.29%	98.36%	99.89%	99.27%	99.72%	99.50%
Number of layers								
2	99.76%	97.79%	99.29%	98.54%	99.87%	99.04%	99.77%	99.41%
3	**99.80%**	**97.79%**	**99.29%**	**98.54%**	**99.90%**	**99.41%**	**99.72%**	**99.56%**
4	99.78%	97.68%	99.29%	98.48%	99.90%	99.27%	99.77%	99.52%
Epoch								
50	99.33%	88.68%	99.76%	93.90%	99.83%	99.22%	99.72%	99.47%
100	99.79%	97.45%	99.29%	98.36%	**99.90%**	**99.41%**	**99.72%**	**99.56%**
200	**99.80%**	**97.79%**	**99.29%**	**98.54%**	99.90%	99.41%	99.72%	99.56%
300	99.80%	97.79%	99.29%	98.54%	99.90%	99.41%	99.72%	99.56%
Optimizer								
Adam	**99.80%**	**97.79%**	**99.29%**	**98.54%**	**99.90%**	**99.41%**	**99.72%**	**99.56%**
Adagrad	99.03%	96.45%	99.41%	97.91%	99.61%	99.04%	99.72%	99.38%
RMSprop	99.78%	97.45%	99.29%	98.36%	99.88%	99.27%	99.72%	99.50%

TABLE 9. The measurement for the interpretability of our framework.

	BUSI Dataset				BUSBRA Dataset			
	Number of k connections				Number of k connections			
	$k=2$	$k=4$	$k=6$	$k=8$	$k=2$	$k=4$	$k=6$	$k=8$
IoU	0.51	0.55	0.61	0.61	0.69	0.76	0.77	0.77
Dice	0.65	0.72	0.85	0.85	0.81	0.86	0.86	0.86

the analysis brings that the $k = 6$ reaches almost the highest performance across various metrics. Thus, the $k = 6$ has been established as the default setting for the following ablation studies.

2) EFFECTIVENESS OF THE PRE-TRAINED MODEL FOR NODE FEATURE EXTRACTION

Node attributes or features directly influence the information that may be convolved and interconnected within a graph. In our research, the formation of node features is mainly based on the extraction from superpixels via pre-trained models. To study the performance of pre-trained model, we conducted a comparative analysis of VGG19, ResNet50, MobileNetV2 and Dense121, as displayed in Table 7. These four models are selected for comparison because they support the minimum size for 32 × 32 of image. In general, the VGG19 pre-trained model achieves the highest contribution of enhancing the features, while Dense121 and MobileNetV2 exhibit relatively low performance.

3) EFFECTIVENESS OF TRAINING PARAMETERS FOR GCN

In order to select the best values for enhancing the training performance, experiments with different training options are conducted on the node-level classification for both datasets. Table 8 shows the various choices and parameters for the training progress. According to these experiments, the result indicates that utilizing three hidden layers gives the highest performance. Besides, the ReLU activation function and Adam optimizer are found to achieve the best results for

TABLE 10. Comparative image-level results between the proposed framework and other methods on BUSI dataset.

Methods	Classes	Accuracy	Precision	Recall	F-Score
Miron et al. [38]	2	86%	-	-	-
Byra [40]	2	95.43%	94.34%	96.15%	95.24%
Zhuang et al. [42]	2	95.48%	98.11%	93.92%	98.33%
Khanna et al. [43]	2	84.9%	-	-	-
Ahila et al. [44]	2	98%	-	98.8%	-
Chowa et al. [51]	2	99.48%	100%	100%	-
Proposed framework (with $k = 6$)	**3**	**98.73%**	**100%**	**97.62%**	**98.80%**

TABLE 11. Comparative image-level results between the proposed framework and other methods on BUSBRA dataset.

Methods	Classes	Accuracy	Precision	Recall	F-Score
ResNet	2	74.67%	62.59%	69.70%	65.95%
VGG19	2	78.40%	72.97%	61.36%	66.67%
MobileNetV2	2	74.40%	69.15%	49.24%	57.52%
EfficientNetB0	2	82.40%	68.97%	90.91%	78.43%
XceptionNet	2	73.33%	61.76%	63.64%	62.69%
Proposed framework (with $k = 6$)	**2**	**98.40%**	**100.00%**	**95.08%**	**97.48%**

both datasets. Only the case of epoch, the BUSI dataset needs 200 epochs to reach the convergence point, while the BUSBRA dataset only requires 100 epochs.

E. INTERPRETABILITY

Understanding the decision-making process is becoming pivotal. This not only plays an essential role in objectively and comprehensively assessing the model but can also assist in explaining and delimiting the final results. In our study, the evaluation of interpretability is employed by comparing the label of nodes with the ground true masks. Since the positions of these nodes are predetermined by the SLIC algorithm, the nodes identified as regions of interest (ROIs) are then compared against the ground truth masks.

To better evaluate this process, Intersection over Union (IoU) and Dice are utilized in our experiments. These metrics are crucial for assessing the overlap of regions, thereby enabling the verification of whether abnormalities fall within the target regions and contributing to the overall result correctly. Thus, this is able to provide a comprehensive perspective on the interpretability of our framework, as shown in Table 9. In general, the IoU and Dice metrics show an upward trend as the number of k connections increases across both datasets. Interestingly, the Dice scores in both datasets also reach the highest value over 0.85. However, in the case of IoU metric, the peak value for BUSI is only 0.61, while BUSBRA dataset gives 0.77. This disparity in IoU performance may be attributed to the limitations in refining regions near the edges or curves by the node-based approach. In contrast, the Dice score aims to emphasize the intersection between regions. Giving the high values in the Dice score means that our framework is able to provide the correct classification with explainability.

F. COMPARED STUDIES

In addition, our comparative analysis offers a careful assessment of the proposed framework against existing state-of-the-art methods. The results depicted in Table 10 demonstrate the performance of our proposed framework compared to prior research conducted on the BUSI dataset. This comparison offers valuable insights into the efficacy of different approaches in handling the complexities of multi-class classification. In spite of the increased complexity of handling three classes, our framework reaches an accuracy of 98.73%. This figure is slightly below the almost perfect accuracy of 99.48% achieved by Chow et al. [51]. Their accuracy is slightly less than that of precision and recall, which reached absolutely 100%. This can be explained by the presence of false negatives or false positives in overall accuracy calculations. Nevertheless, our result is still considered exceptionally high, especially considering the inclusion of an additional class in this study. Precision in the proposed framework is maintained at a flawless value of 100%, which is equivalent to the highest performance achieved by Chow et al. [51] and exceeds the figures of other approaches, including those of Byra [40] and Zhuang et al. [42] - at 98.11% and 94.34%, respectively. Additionally, the recall of 97.62% and F-score of 98.80% are proven as the highest value compared to other studies, which indices the robustness of model in a more challenging three-class classification scenario. Besides, the comparative results for the BUSBRA dataset are shown in Table 11. To the best of our knowledge, this dataset has not been utilized

in other studies for evaluation purposes. Thus, our study compares with the baseline CNN pre-trained models. Despite the highest accuracy result of EfficientNetB0 reaching over 82%, the remaining pre-trained models still stay below 80% for all metrics. Our result indicates a superior performance compared to other models, which achieves over 95% for the mentioned evaluation metrics.

V. CONCLUSION

In this study, we have introduced a novel node-based GCN framework for classifying breast cancer based on ultrasound imaging. Instead of handling the conventional image data, the conversion to graph data is employed to capture more useful information. Subsequently, our approach leverages the capabilities of GCN to process these graph structure data, enabling the effective classification of superpixel nodes derived from ultrasound images into benign, malignant, or normal categories. The findings of our experiment, which were assessed using two separate datasets, clearly show that our method outperforms both traditional and other state-of-the-art deep learning techniques - achieving the highest accuracy of 98.73% for BUSI dataset and 98.40% for BUS-BRA dataset. This demonstrates the potential of node-based GCN to give robustness in classification and perhaps enhance the effectiveness of early breast cancer detection. Although the proposed node-based approach achieved an enhanced performance in classification results, localizing nodes at the tumor boundary still cannot be perfectly delineated. This limitation may affect the cases of small tumors, which are more susceptible to misclassification. Future research efforts will focus on improving the precision at the tumor edge by refining the detailed boundary of the nodes, which could also enhance the interpretability of the results accurately. Moreover, expanding the current research to adopt 3D medical imaging approaches such as CT or MRI could describe the volumetric spatial relationships inherent in actual cases. This may integrate with additional information for the node features and encapsulate three-dimensional spatial context effectively. Thus, this future work is significant for providing more reliable support in clinical decision-making processes.

REFERENCES

[1] A. G. Waks and E. P. Winer, "Breast cancer treatment: A review," *JAMA*, vol. 321, no. 3, p. 288, Jan. 2019.

[2] (2019). *Breast Cancer Statistics, 2019—DeSantis—2019—CA: A Cancer Journal for Clinicians*. Accessed: Apr. 4, 2024. [Online]. Available: https://acsjournals.onlinelibrary.wiley.com/doi/10.3322/caac.21583

[3] *Global Breast Cancer Initiative Implementation Framework: Assessing, Strengthening and Scaling Up of Services for the Early Detection and Management of Breast Cancer: Executive Summary*. Accessed: Apr. 25, 2024. [Online]. Available: https://www.who.int/publications

[4] Kriti, J. Virmani, and R. Agarwal, "Deep feature extraction and classification of breast ultrasound images," *Multimedia Tools Appl.*, vol. 79, nos. 37–38, pp. 27257–27292, Oct. 2020.

[5] S. Hofvind, B. F. Iversen, L. Eriksen, B. M. Styr, K. Kjellevold, and K. D. Kurz, "Mammographic morphology and distribution of calcifications in ductal carcinoma in situ diagnosed in organized screening," *Acta Radiologica*, vol. 52, no. 5, pp. 481–487, Jun. 2011.

[6] Y. V. Nalawade, "Evaluation of breast calcifications," *Indian J. Radiol. Imag.*, vol. 19, no. 4, pp. 282–286, Oct. 2009, doi: 10.4103/0971-3026.57208.

[7] L. Tabār, "Beyond randomized controlled trials," *Cancer*, vol. 91, no. 9, pp. 1724–1731, 2001.

[8] B. E. Bejnordi, "Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer," *JAMA*, vol. 318, no. 22, p. 2199.

[9] L. Sun, M. Qi, X. Cui, and Q. Song, "The clinical application of combined ultrasound, mammography, and tumor markers in screening breast cancer among high-risk women," *Comput. Math. Methods Med.*, vol. 2022, pp. 1–6, Jul. 2022, doi: 10.1155/2022/4074628.

[10] P. Skaane, A. I. Bandos, L. T. Niklason, S. Sebuødegard, B. H. Østeras, R. Gullien, D. Gur, and S. Hofvind, "Digital mammography versus digital mammography plus tomosynthesis in breast cancer screening: The Oslo tomosynthesis screening trial," *Radiology*, vol. 291, no. 1, pp. 23–30, Apr. 2019.

[11] Q. Dan, T. Zheng, L. Liu, D. Sun, and Y. Chen, "Ultrasound for breast cancer screening in resource-limited settings: Current practice and future directions," *Cancers*, vol. 15, no. 7, p. 2112, Mar. 2023.

[12] G. Chen, Y. Dai, and J. Zhang, "RRCNet: Refinement residual convolutional network for breast ultrasound images segmentation," *Eng. Appl. Artif. Intell.*, vol. 117, Jan. 2023, Art. no. 105601.

[13] Y. Zhou, H. Chen, Y. Li, Q. Liu, X. Xu, S. Wang, P.-T. Yap, and D. Shen, "Multi-task learning for segmentation and classification of tumors in 3D automated breast ultrasound images," *Med. Image Anal.*, vol. 70, May 2021, Art. no. 101918.

[14] K. Drukker, N. P. Gruszauskas, C. A. Sennett, and M. L. Giger, "Breast U.S. computer-aided diagnosis workstation: Performance with a large clinical diagnostic population," *Radiology*, vol. 248, no. 2, pp. 392–397, Aug. 2008.

[15] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778, doi: 10.1109/CVPR.2016.90.

[16] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[17] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, "MobileNets: Efficient convolutional neural networks for mobile vision applications," 2017, *arXiv:1704.04861*.

[18] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," 2019, *arXiv:1905.11946*.

[19] M. Y. Ansari, "Towards developing a lightweight neural network for liver CT segmentation," in *Lecture Notes in Electrical Engineering*. Cham, Switzerland: Springer, 2023, pp. 27–35.

[20] M. Y. Ansari, I. A. C. Mangalote, D. Masri, and S. P. Dakua, "Neural network-based fast liver ultrasound image segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2023.

[21] M. Y. Ansari, I. A. Changaai Mangalote, P. K. Meher, O. Aboumarzouk, A. Al-Ansari, O. Halabi, and S. P. Dakua, "Advancements in deep learning for B-mode ultrasound segmentation: A comprehensive review," *IEEE Trans. Emerg. Topics Comput. Intell.*, vol. 8, no. 3, pp. 2126–2149, Jun. 2024, doi: 10.1109/TETCI.2024.3377676.

[22] F. Ullah, M. Nadeem, M. Abrar, F. Amin, A. Salam, A. Alabrah, and H. AlSalman, "Evolutionary model for brain cancer-grading and classification," *IEEE Access*, vol. 11, pp. 126182–126194, 2023.

[23] M. F. Almufareh, M. Imran, A. Khan, M. Humayun, and M. Asim, "Automated brain tumor segmentation and classification in MRI using YOLO-based deep learning," *IEEE Access*, vol. 12, pp. 16189–16207, 2024, doi: 10.1109/ACCESS.2024.3359418.

[24] P. Rai, M. Y. Ansari, M. Warfa, H. Al-Hamar, J. Abinahed, A. Barah, S. P. Dakua, and S. Balakrishnan, "Efficacy of fusion imaging for immediate post-ablation assessment of malignant liver neoplasms: A systematic review," *Cancer Med.*, vol. 12, no. 13, pp. 14225–14251, Jul. 2023.

[25] M. Y. Ansari, M. Qaraqe, R. Righetti, E. Serpedin, and K. Qaraqe, "Unveiling the future of breast cancer assessment: A critical review on generative adversarial networks in elastography ultrasound," *Frontiers Oncol.*, vol. 13, pp. 1–26, Dec. 2023, doi: 10.3389/FONC.2023.1282536.

[26] M. Alruwaili and W. Gouda, "Automated breast cancer detection models based on transfer learning," *Sensors*, vol. 22, no. 3, p. 876, Jan. 2022.

[27] Z. Hussain, F. Gimenez, D. Yi, and D. Rubin, "Differential data augmentation techniques for medical imaging classification tasks," in *AMIA Annu. Symp. Proc.*, vol. 2017, Apr. 2018, pp. 979–984.

[28] L. Wang, M. Zhang, G. He, D. Shen, and M. Meng, "Classification of breast lesions on DCE-MRI data using a fine-tuned MobileNet," *Diagnostics*, vol. 13, no. 6, p. 1067, Mar. 2023.

[29] S. Gengtian, B. Bing, and Z. Guoyou, "EfficientNet-based deep learning approach for breast cancer detection with mammography images," in *Proc. 8th Int. Conf. Comput. Commun. Syst. (ICCCS)*, Apr. 2023, pp. 972–977, doi: 10.1109/ICCCS57501.2023.10151156.

[30] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: A review of methods and applications," *AI Open*, vol. 1, pp. 57–81, Jan. 2020.

[31] Y. Sang and W. Li, "Classification study of Alzheimer's disease based on self-attention mechanism and DTI imaging using GCN," *IEEE Access*, vol. 12, pp. 24387–24395, 2024.

[32] R. Li, S. Wang, F. Zhu, and J. Huang, "Adaptive graph convolutional neural networks," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 3546–3553.

[33] F. Liu, X. Qian, L. Jiao, X. Zhang, L. Li, and Y. Cui, "Contrastive learning-based dual dynamic GCN for SAR image scene classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1–15, Jan. 2022, doi: 10.1109/TNNLS.2022.3174873.

[34] S. Jia, S. Jiang, S. Zhang, M. Xu, and X. Jia, "Graph-in-graph convolutional network for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 1, pp. 1–15, Jan. 2022, doi: 10.1109/TNNLS.2022.3182715.

[35] S. Chai, R. K. Jain, Y. Li, J. Liu, T. Tateyama, and Y.-W. Chen, "Adaptive graph convolutional networks for medical image segmentation," in *Proc. 45th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2023, pp. 1–4.

[36] P. H. C. Avelar, A. R. Tavares, T. L. T. da Silveira, C. R. Jung, and L. C. Lamb, "Superpixel image classification with graph attention networks," in *Proc. 33rd SIBGRAPI Conf. Graph., Patterns Images (SIBGRAPI)*, Nov. 2020, pp. 203–209.

[37] L. Ismaila, P. Rasti, J.-M. Lemée, and D. Rousseau, "Toward more frugal models for functional cerebral networks automatic recognition with resting-state fMRI," 2023, *arXiv:2307.01953*.

[38] M. Miron, S. Moldovanu, and A. L. Culea-Florescu, "A multi-layer feed forward neural network for breast cancer diagnosis from ultrasound images," in *Proc. 26th Int. Conf. Syst. Theory, Control Comput. (ICSTCC)*, Moldova, Oct. 2022, pp. 421–425.

[39] H. Yu, H. Sun, J. Li, L. Shi, N. Bao, H. Li, W. Qian, and S. Zhou, "Effective diagnostic model construction based on discriminative breast ultrasound image regions using deep feature extraction," *Med. Phys.*, vol. 48, no. 6, pp. 2920–2928, Jun. 2021.

[40] M. Byra, "Breast mass classification with transfer learning based on scaling of deep representations," *Biomed. Signal Process. Control*, vol. 69, Aug. 2021, Art. no. 102828.

[41] G. Ayana, J. Park, J.-W. Jeong, and S.-W. Choe, "A novel multistage transfer learning for ultrasound breast cancer image classification," *Diagnostics*, vol. 12, no. 1, p. 135, Jan. 2022.

[42] Z. Zhuang, Z. Yang, A. N. J. Raj, C. Wei, P. Jin, and S. Zhuang, "Breast ultrasound tumor image classification using image decomposition and fusion based on adaptive multi-model spatial feature fusion," *Comput. Methods Programs Biomed.*, vol. 208, Sep. 2021, Art. no. 106221.

[43] P. Khanna, M. Sahu, and B. Kumar Singh, "Improving the classification performance of breast ultrasound image using deep learning and optimization algorithm," in *Proc. IEEE Int. Conf. Technol., Res., Innov. Betterment Soc. (TRIBES)*, Dec. 2021, pp. 1–6.

[44] A. A, P. M, S. Bourouis, S. S. Band, A. Mosavi, S. Agrawal, and M. Hamdi, "Meta-heuristic algorithm-tuned neural network for breast cancer diagnosis using ultrasound images," *Frontiers Oncol.*, vol. 12, pp. 1–26, Jun. 2022.

[45] J. Wang, "Auto-weighting for breast cancer classification in multimodal ultrasound," in *Medical Image Computing and Computer Assisted Intervention—MICCAI*. Cham, Switzerland: Springer, 2020, pp. 190–199.

[46] J. Song, Y. Zheng, M. Zakir Ullah, J. Wang, Y. Jiang, C. Xu, Z. Zou, and G. Ding, "Multiview multimodal network for breast cancer diagnosis in contrast-enhanced spectral mammography images," *Int. J. Comput. Assist. Radiol. Surgery*, vol. 16, no. 6, pp. 979–988, Jun. 2021.

[47] K. Atrey, B. K. Singh, and N. K. Bodhey, "Multimodal classification of breast cancer using feature level fusion of mammogram and ultrasound images in machine learning paradigm," *Multimedia Tools Appl.*, vol. 83, no. 7, pp. 21347–21368, Aug. 2023.

[48] K. Atrey, B. K. Singh, N. K. Bodhey, and R. Bilas Pachori, "Mammography and ultrasound based dual modality classification of breast cancer using a hybrid deep learning approach," *Biomed. Signal Process. Control*, vol. 86, Sep. 2023, Art. no. 104919.

[49] M. Ibrahim, S. Henna, and G. Cullen, "Multi-graph convolutional neural network for breast cancer multi-task classification," in *Communications in Computer and Information Science*. Cham, Switzerland: Springer, 2023, pp. 40–54.

[50] L. T. Duong, C. Q. Chu, P. T. Nguyen, S. T. Nguyen, and B. Q. Tran, "Edge detection and graph neural networks to classify mammograms: A case study with a dataset from Vietnamese patients," *Appl. Soft Comput.*, vol. 134, Feb. 2023, Art. no. 109974.

[51] S. S. Chowa, S. Azam, S. Montaha, I. J. Payel, M. R. I. Bhuiyan, M. Z. Hasan, and M. Jonkman, "Graph neural network-based breast cancer diagnosis using ultrasound images with optimized graph construction integrating the medically significant features," *J. Cancer Res. Clin. Oncol.*, vol. 149, no. 20, pp. 18039–18064, Dec. 2023.

[52] W. Al-Dhabyani, M. Gomaa, H. Khaled, and A. Fahmy, "Dataset of breast ultrasound images," *Data Brief*, vol. 28, Feb. 2020, Art. no. 104863.

[53] W. Gómez-Flores, M. J. Gregorio-Calas, and W. Coelho de Albuquerque Pereira, "BUS-BRA: A breast ultrasound dataset for assessing computer-aided diagnosis systems," *Med. Phys.*, vol. 51, no. 4, pp. 3110–3123, Apr. 2024.

[54] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Susstrunk, "SLIC superpixels compared to State-of-the-Art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012, doi: 10.1109/TPAMI.2012.120.



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